# Astley Santos

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## PROFILE

Software Engineer with a Master's in Computer Science specialised in Artificial Intelligence. Experienced in building end-to-end systems spanning from backend, frontend, and machine learning pipelines, with a strong focus on scalable architecture and research-driven problem solving.

Proven ability to design and implement complex software systems and apply Machine Learning techniques to real-world engineering and scientific problems.

Seeking to contribute to impactful products at the intersections of software engineering and/or AI.

## TECHNICAL SKILLS

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| **Programming Languages** | Java, Python, JavaScript, TypeScript, SQL |
| **AI & Machine Learning** | PyTorch, Scikit-learn, Pandas, NumPy and other related libraries |
| **Software Development** | React.js, Spring Boot, Angular, Android, Node.js, REST APIs |
| **Misc** | Docker, Ansible, Gitlab CI/CD, Arduino |

## WORK EXPERIENCE

**Software Engineer Intern, Yuri GmbH – 6 months**
**07/2025 – 12/2025**

- Built the foundational architecture and MVP of Yuri's Mission Control System using Java/JavaFX, for planning, monitoring and managing space-based biotechnology experiments
- Developed frontend and backend logic for telemetry visualisation and telecommand execution, supporting real-time control of payloads and experiment workflows
- Defined modular architecture and scalable design patterns to support future mission expansion
- Collaborated in a high-stakes, research-driven environment bridging software engineering and space systems

**Teaching Assistant and Software Developer – University of Luxembourg (incl. LCSB)**
**09/2022 – 01/2024**

- Contributed to feasibility analysis and early-stage development of a mobile health application for Parkinson's monitoring and biomedical research for the Luxembourg Centre for Systems Biomedicine (LCSB)
- Supported cross-platform prototyping (iOS/watchOS and Android/WearOS) to evaluate technical approaches and system capabilities
- Collaborated with researchers in a multidisciplinary environment bridging software development and biomedical applications
- Designed and automated testing workflows for grading programming assignments in a Web Development course
- Improved efficiency and consistency of evaluation through structured test pipelines and tooling

# PROJECTS

**Affective Computing & GUI Evaluation – Master's Thesis**

- Designed and implemented an EEG-based affective computing system to classify user emotional responses to GUI designs
- Built end-to-end ML pipeline: data processing, feature extraction, model training, and evaluation
- Applied Machine Learning techniques (SVM, Random Forest, KNN) and Deep Learning models (CNN-based architectures) for classification tasks.
- Optimised model performance through hyperparameter tuning, and comparative evaluation across architectures.

**Electrical Vehicle Taxi Optimisation – Genetic Algorithm**

- Developed a Genetic Algorithm to optimise routing and scheduling of electrical vehicles taxis
- Applied multi-objective optimisation techniques to balance efficiency, energy consumption, and operational constraints
- Collaborated with doctoral researchers on a real-world research problem and produced a formal research report

**Automation Pipeline – DevOps**

- Designed and implemented a CI/CD pipeline to automate environment setup, application deployment, and testing workflows.
- Set up infrastructure automation using Docker, Ansible, and Vagrant across development, staging and production environments, improving deployment consistency and reducing manual configuration through infrastructure-as-code practices

**Autonomous Vehicle Simulation**

- Developed an autonomous driving simulation using ROS and Docker in the Udacity self-driving car environment.
- Implemented core functionalities including vehicle control (acceleration, braking), path planning, and perception
- Built a computer vision module for traffic light detection and classification

# ADDITIONAL

**Amazon Student Programs x University of Luxembourg: All to Tech Mentorship Program**
**05/2024 – 09/2024**

- Selected for a technical mentorship program focused on Software Engineering interview preparations. Completed multiple mock interviews covering data structures and algorithms, and problem-solving techniques.

**Organization of a Workshop – PC Building**
**2021**

- Co-organised and delivered a workshop introducing PC hardware components and system assembly, guiding the participants through the process of building a computer from scratch.
- Promoted technical curiosity and hands-on learning within the student community

# EDUCATION

University of Luxembourg
Luxembourg
**2021-2025**

**Master's in Information and Computer Sciences**

- Profiles: Artificial Intelligence and Software Engineering

Universidade Lusófona de Humanidades e Tecnologias
Portugal
**2017-2020**

**Bachelor's in Computer Engineering**

- Main courses: Algorithms and Data Structures, Artificial Intelligence, Data Science, Programming Languages, Computer Architecture, Operating Systems, Databases, Software Engineering, and many others.

# PERSONAL

**Languages** Portuguese (Native), English (Proficient), French (B2), German (A2)